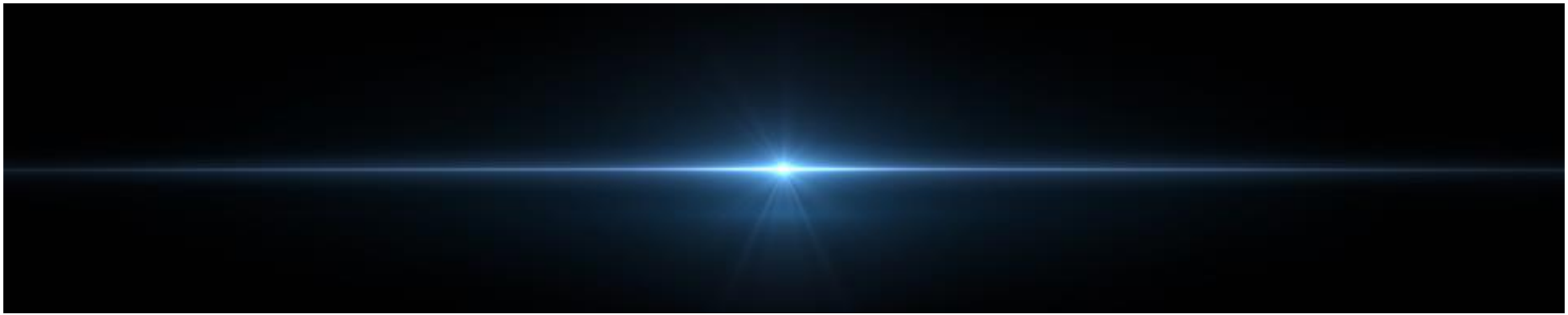
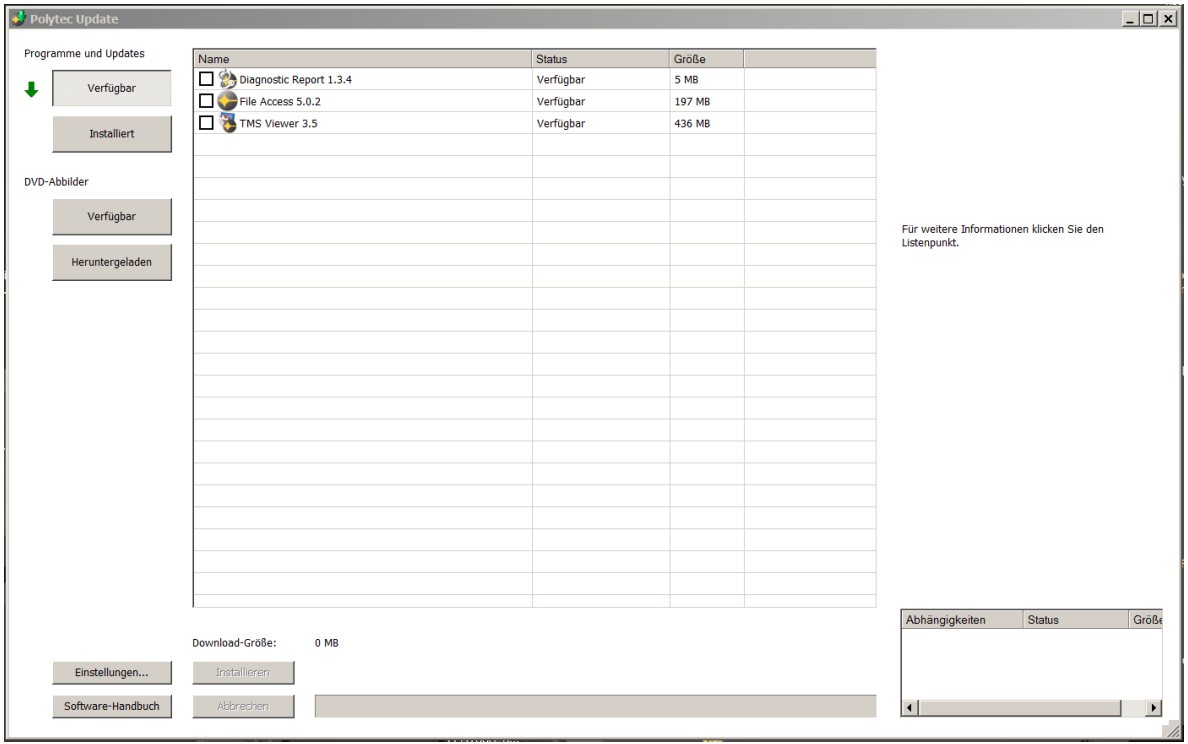


Polytec File Access – Bsp. Matlab, Excel

Polytec File Access



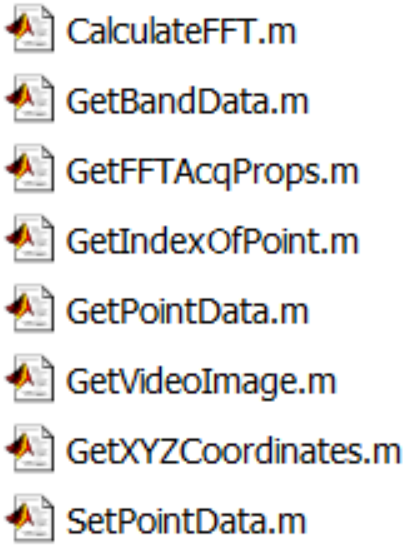
- Kein Dongle erforderlich
- Direkter Zugriff auf .pvd und .svd Dateien (z.B. mit Matlab)

- Über „Polytec-Update“ verfügbar

https://www.polytec.com/de/vibrometrie/produkte/software/polytec-update

Polytec File Access - Matlab

C:\Program Files\Polytec\Polytec File Access 5.6\Examples\Matlab



Notwendige Skripte ins entsprechende Arbeitsverzeichnis kopieren bzw. richtigen Pfad in Matlab auswählen

Polytec File Access - Matlab

C:\Program Files\Polytec\Polytec File Access 5.6\Examples\Matlab

```
% [x,y,usd] = GetPointData(filename, domainname, channelname, signalname, displayname,
%   point, frame)
%
% Gets original or user defined data from a polytec file.
%
% filename is the path of the .pvd or .svd file
% domainname is the name of the domain, e.g. 'FFT' or 'Time'
% channelname is the name of the channel, e.g. 'Vib' or 'Ref1' or 'Vib &
%   Ref1' or 'Vib X' or 'Vib Y' or 'Vib Z'.
% signalname is the name of the signal, e.g. 'Velocity' or 'Displacement'
% displayname is the name of the display, e.g. 'Real' or 'Magnitude' or
%   'Samples'. If the display name is 'Real & Imag.' the data is returned
%   as complex values.
% point is the (1-based) index of the point to get data from. If point is
%   0 the data of all points will be returned. y will contain the data of
%   point i at row index i.
% frame is the frame number of the data. for data acquired in MultiFrame
%   mode, 0 is the averaged frame and 1-n are the other frames. For user
%   defined datasets the frame number is in the range 1-n where n is the
%   number of frames in the user defined dataset. For all other data,
%   use frame number 0.
%
% returns x, the x axis values of the data
% returns y, the data. colomns correspond to the x-axis, rows to the point
%   index. for point = 0: rows for points that have no data are set to zeros.
% returns usd, a struct describing the signal
%
function [x,y,usd] = GetPointData(filename, domainname, channelname, signalname, displayname, point, frame)
%
file = actxserver('PolyFile.PolyFile');
try
    file.Open(filename);
```

Beschreibung als Kommentar im Skript!

Polytec File Access - Matlab

GetPointData.m

[x,y,usd] = GetPointData(‘filename’, ‘domainname’, ‘channelname’, ‘signalname’, ‘displayname’, point, frame)

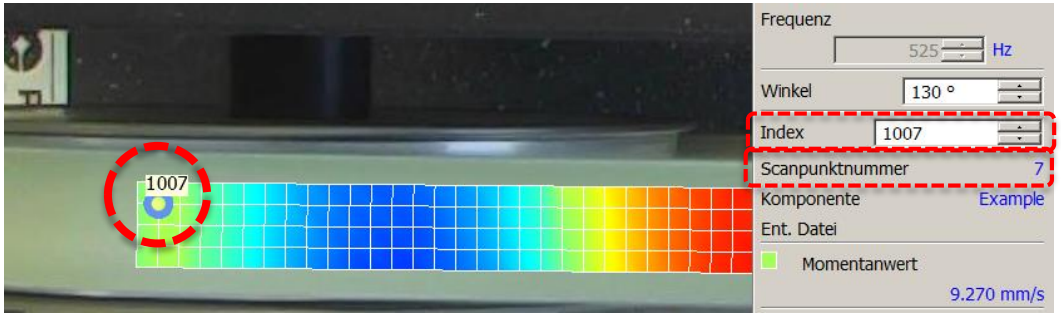
Pfad .svd .pvd	FFT Time	Vib Vib X Vib Y Vib Z Ref1 . .	Velocity Displacement Acceleration . .	Magnitude Real . . **Samples** **(Zeitdaten)**	Indexnr. **0** alle Punkte	0

Groß/ Kleinschreibung beachten!

Deutsche Schreibweise geht auch

Polytec File Access - Matlab

GetPointData.m



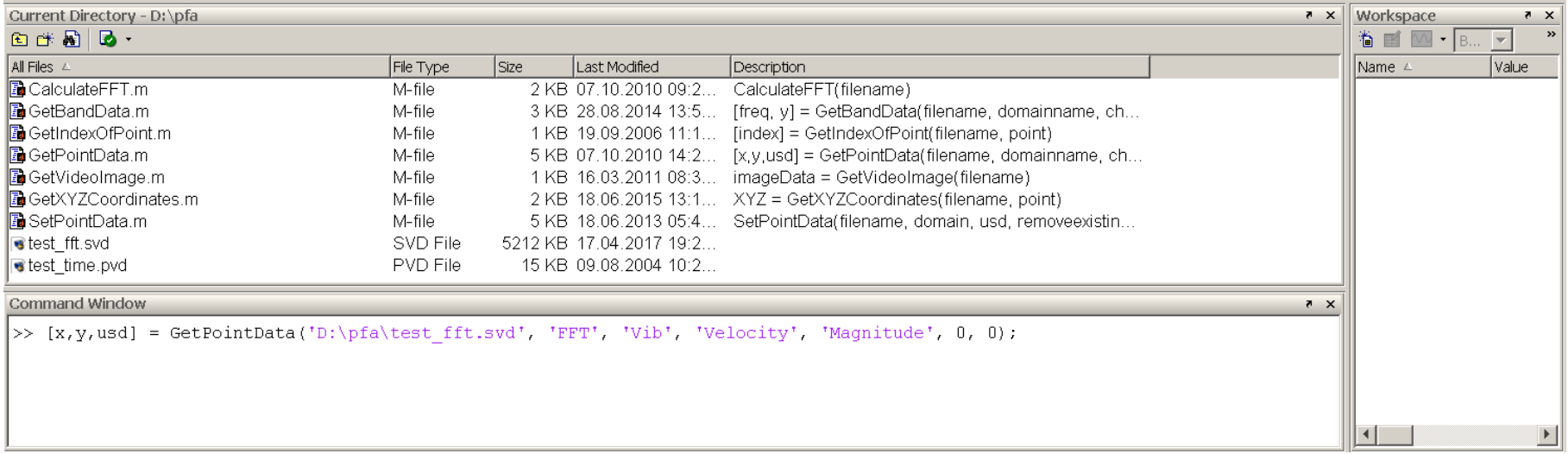
ursprünglicher Index

Vorsicht bei geänderten Punkt-Indizes (PSV „Indizes ändern“)!

Im Skript wird die Scanpunktnummer (ursprünglicher Index) angegeben.

Polytec File Access - Matlab

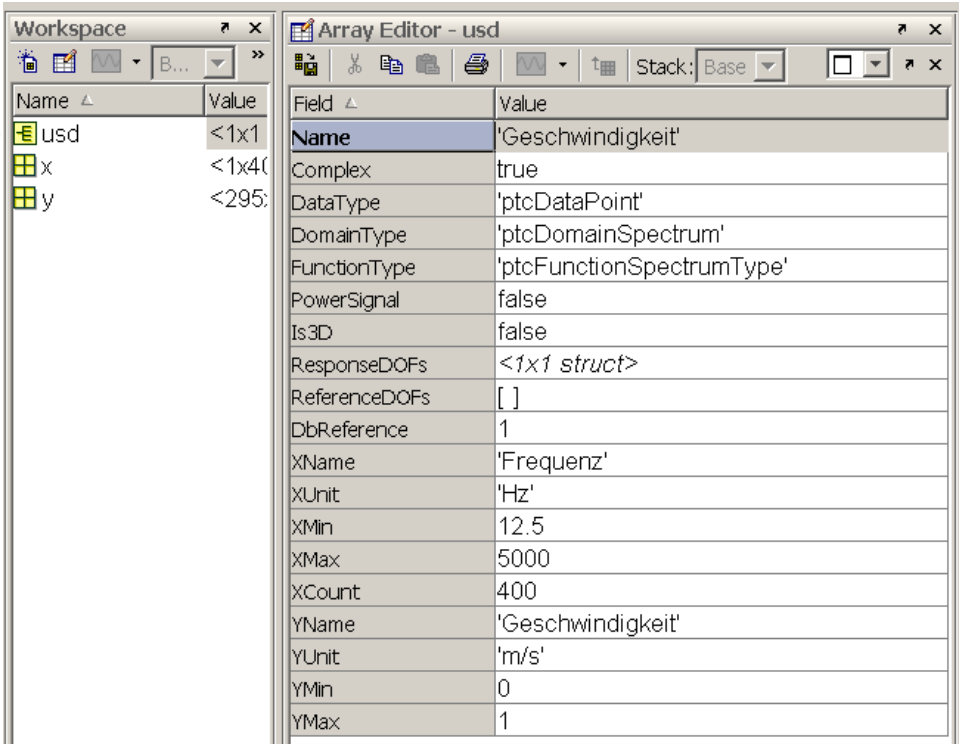
GetPointData.m (Bsp. FFT, alle Punkte)



Polytec File Access - Matlab

GetPointData.m

Beschreibung



Polytec File Access - Matlab

GetPointData.m (Bsp. FFT, alle Punkte)

Frequenz

Daten

Workspace

Stack: Base

Name	Value	Min
usd	<1x1 struct>	
x	<1x400 double>	12.5
y	<295x400 double>	0.0002

Array Editor - y

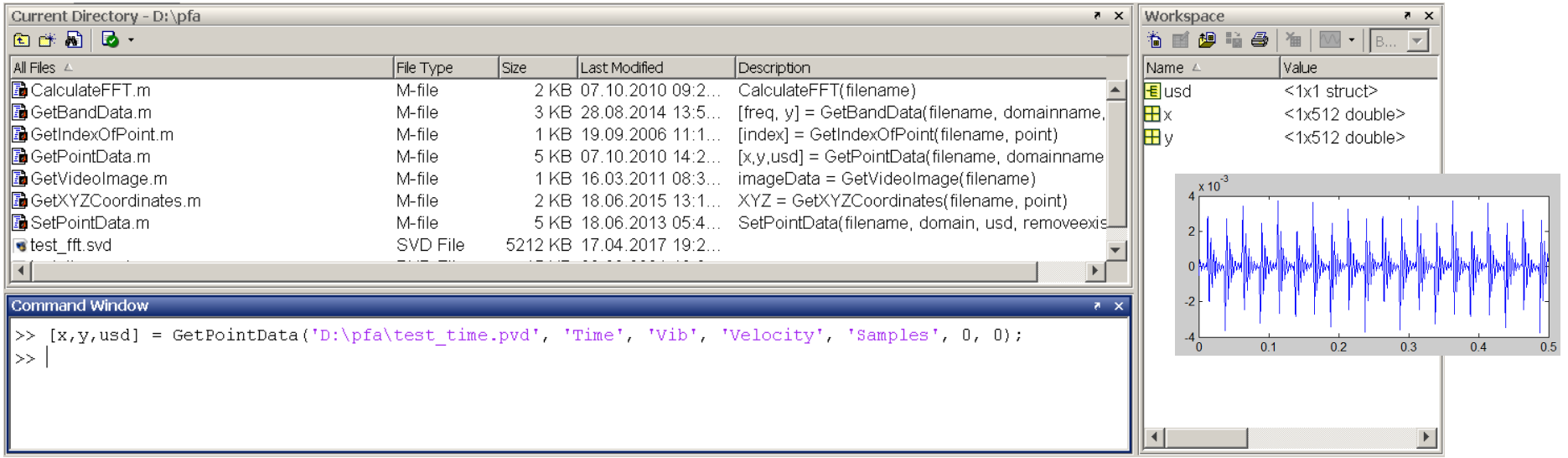
Stack: Base

	1	2	3	4	5	6	7	8	9	10	11
1	0.0035	0.0038	0.0017	0.0014	0.0009	0.0007	0.0005	0.0005	0.0006	0.0005	0.0
2	0.0036	0.004	0.002	0.0017	0.001	0.0008	0.0006	0.0006	0.0006	0.0005	0.0
3	0.0036	0.0039	0.0019	0.0017	0.001	0.0007	0.0006	0.0005	0.0005	0.0005	0.0
4	0.0034	0.0037	0.0019	0.0016	0.0008	0.0008	0.0006	0.0006	0.0005	0.0005	0.0
5	0.0037	0.0039	0.0019	0.0017	0.0009	0.0007	0.0007	0.0006	0.0005	0.0005	0.0
6	0.0033	0.0034	0.0017	0.0018	0.0013	0.0009	0.0007	0.0008	0.0008	0.0008	0.0
7	0.0032	Amplitude in m/s		19	0.0012	0.0008	0.0007	0.0007	0.0006	0.0007	0.
8	0.0033	0.0035	0.0019	0.002	0.0014	0.0009	0.0007	0.0007	0.0007	0.0007	0.0
9	0.0033	0.0035	0.0017	0.0018	0.0012	0.0009	0.0007	0.0007	0.0007	0.0007	0.0
10	0.0032	0.0033	0.0019	0.002	0.0013	0.0009	0.0007	0.0007	0.0007	0.0007	0.0
11	0.0034	0.0033	0.0021	0.0025	0.0019	0.0012	0.0012	0.0011	0.0011	0.0011	0.0
12	0.0033	0.0033	0.0022	0.0027	0.0021	0.0012	0.0014	0.0014	0.0013	0.0011	0.0
13	0.0034	0.0033	0.0021	0.0025	0.002	0.0013	0.0014	0.0012	0.0012	0.0011	0.0
14	0.0034	0.0033	0.0021	0.0024	0.002	0.0011	0.0013	0.0013	0.0012	0.0011	0.0
15	0.004	0.0035	0.0023	0.0025	0.002	0.0012	0.0012	0.0013	0.0011	0.0012	0.0
	0.0036	0.0034	0.0022	0.0029	0.0023	0.0015	0.0016	0.0014	0.0017	0.0017	0.0

Index

Polytec File Access - Matlab

GetPointData.m (Bsp. Zeit, 1-Punkt-Daten)



Polytec File Access - Matlab

GetVideoImage.m

Current Directory - D:\pfa

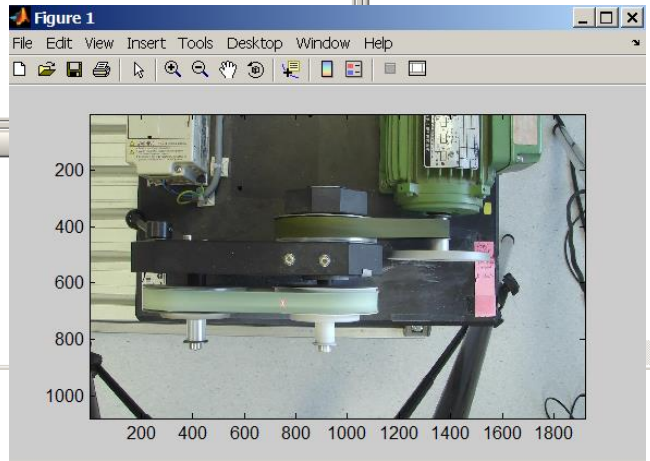
All Files	File Type	Size	Last Modified	Description
CalculateFFT.m	M-file	2 KB	07.10.2010 09:2...	CalculateFFT(filename)
GetBandData.m	M-file	3 KB	28.08.2014 13:5...	[freq, y] = GetBandData(filename, domainname, ch...
GetIndexOfPoint.m	M-file	1 KB	19.09.2006 11:1...	[index] = GetIndexOfPoint(filename, point)
GetPointData.m	M-file	5 KB	07.10.2010 14:2...	[x,y,usd] = GetPointData(filename, domainname, ch...
GetVideoImage.m	M-file	1 KB	16.03.2011 08:3...	imageData = GetVideoImage(filename)
GetXYZCoordinates.m	M-file	2 KB	18.06.2015 13:1...	XYZ = GetXYZCoordinates(filename, point)
SetPointData.m	M-file	5 KB	18.06.2013 05:4...	SetPointData(filename, domain, usd, removeexistin...
test_fft.svd	SVD File	5212 KB	17.04.2017 19:2...	
test_time.pvd	PVD File	15 KB	09.08.2004 10:2...	

Workspace

Name	Value
imageData	<1080x1920x3 uint8>

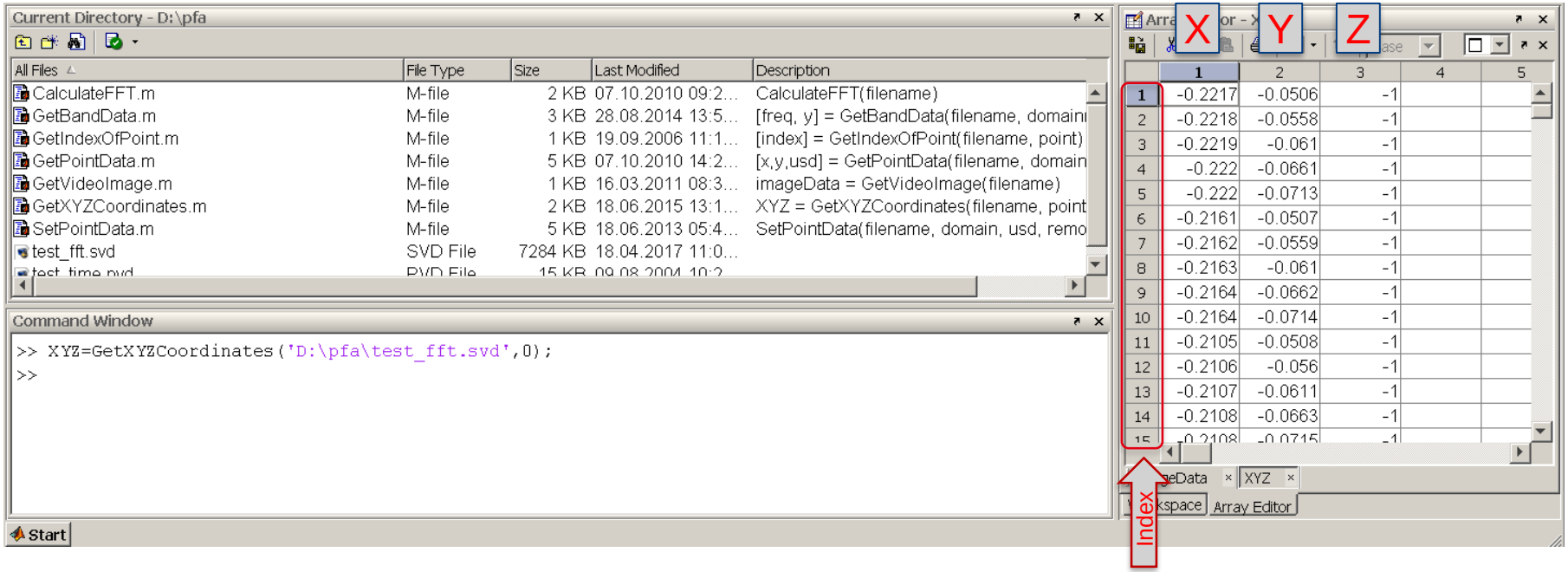
Command Window

```
>> imageData = GetVideoImage('D:\pfa\test_fft.svd');
>> image(imageData)
>>
```



Polytec File Access - Matlab

GetXYZCoordinates.m



Polytec File Access - Excel

C:\Program Files\Polytec\Polytec File Access 5.6\Examples\Excel

PolyFile.xls

